

A VIRTUAL EVENT

5 DAYS OF STEM CELLS

Connect. Discover. Advance.

Join us for the world's leading virtual stem cell event.

gibco

Stay on the leading edge of stem cell research

The Gibco™ 5 Days of Stem Cells virtual event connects you to the latest stem cell techniques, research breakthroughs, and esteemed scientists from around the world—all from the comfort of anywhere

We'll share developments, discoveries, and cutting-edge content connected to a wide variety of stem cell applications, including disease modeling, cell and gene therapy, 3D modeling, and much more.

The 5-day virtual agenda is packed full of incredible insights in the form of:

- Leading scientific presentations from thought leaders around the world
- Behind-the-scenes virtual training demos
- Scientific poster sessions
- Hundreds of key stem cell tools and resources
- A global network of researchers including our stem cell experts and technical support

It's all happening October 12–16, 2020.



Register for free at thermofisher.com/5daysofstemcells

ThermoFisher
SCIENTIFIC

Advances in Inline Quantification of Co-Eluting Proteins in Chromatography: Process-Data-Based Model Calibration and Application Towards Real-Life Separation Issues

Nina Brestrich,¹ Adrian Sanden,¹ Axel Kraft,¹ Karl McCann,² Joseph Bertolini,² Jürgen Hubbuch¹

¹Karlsruhe Institute of Technology, Engler-Bunte-Ring 1 76131, Karlsruhe, Germany; telephone: +49 721 608 42557; fax: +49 721 608 46240; e-mail: juergen.hubbuch@kit.edu

²CSL Behring Australia, Broadmeadows, VIC, Australia

ABSTRACT: Pooling decisions in preparative liquid chromatography for protein purification are usually based on univariate UV absorption measurements that are not able to differentiate between product and co-eluting contaminants. This can result in inconsistent pool purities or yields, if there is a batch-to-batch variability of the feedstock. To overcome this analytical bottleneck, a tool for selective inline quantification of co-eluting model proteins using mid-UV absorption spectra and Partial Least Squares Regression (PLS) was presented in a previous study and applied for real-time pooling decisions. In this paper, a process-data-based method for the PLS model calibration will be introduced that allows the application of the tool towards chromatography steps of real-life processes. The process-data-based calibration method uses recorded inline mid-UV absorption spectra that are correlated with offline fraction analytics to calibrate PLS models. In order to generate average spectra from the inline data, a Visual Basic for Application macro was successfully developed. The process-data-based model calibration was established using a ternary model protein system. Afterwards, it was successfully demonstrated in two case studies that the calibration method is applicable towards real-life separation issues. The calibrated PLS models allowed a successful quantification of the co-eluting species in a cation-exchange-based aggregate and fraction removal during the purification of monoclonal antibodies and of co-eluting serum proteins in an anion-exchange-based purification of Cohn supernatant I. Consequently, the presented process-data-based PLS model calibration in combination with the tool for selective inline quantification has a great potential for the monitoring of future chromatography steps and may contribute to manage batch-to-batch variability by real-time pooling decisions.

Biotechnol. Bioeng. 2015;112: 1406–1416.

© 2015 Wiley Periodicals, Inc.

KEYWORDS: process analytical technology; inline monitoring; chemometrics; partial least squares regression; selective protein quantification; protein analytics; bioprocess monitoring

Introduction

In the production of biopharmaceuticals, a consistent product quality is of high importance. A key issue compromising a consistent quality is batch-to-batch variability. In order to manage variability and to ensure a consistent product quality, the US Food and Drug Administration's (FDA's) Process Analytical Technology (PAT) initiative promotes the monitoring and control of critical quality attributes and process parameters as well as the use of modern, multivariate data acquisition tools (FDA, 2004). Especially in liquid chromatography, the main workhorse in biopharmaceutical protein purification, batch-to-batch variability represents a major issue. Typical reasons are aging of columns, variations in the composition of buffers and feed-concentrations, or lot-to-lot variability of resins, which may result in different retention times or peak shapes of the eluting proteins. This makes the decision on when to start or to stop the pooling of the target fraction a great challenge, especially in the case of co-eluting proteins.

Pooling decisions in preparative chromatography are usually based on univariate UV absorption measurements, as this approach is easy to implement (Follman and Fahrner, 2004; Zhou et al., 2007). However, univariate UV absorption measurements do not allow a differentiation between co-eluting proteins. This is especially disadvantageous during polishing steps that aim to remove product-related impurities. These impurities exhibit similar physicochemical properties as the product, which usually results in a poor resolution. In these cases, pooling based on univariate UV absorption results in inconsistent pool purities or yields, if there is a batch-to-batch variability of the feedstock.

In order to tackle this issue, various PAT tools for chromatography have been developed in the last years that enable the differentiation between product and co-eluting contaminants

Correspondence to: Jürgen Hubbuch

Contract grant sponsor: Ministry of Science, Research, and the Arts of Baden-Württemberg, Germany

Contract grant number: Az.: 33-7533-7-11.6-2

Received 11 September 2014; Revision received 8 December 2014; Accepted 5 January 2015

Accepted manuscript online 12 February 2015;

Article first published online 23 April 2015 in Wiley Online Library (<http://onlinelibrary.wiley.com/doi/10.1002/bit.25546/abstract>).

DOI 10.1002/bit.25546

(Fahrner et al., 1998; Kaltenbrunner et al., 2012; Mendhe et al., 2014; Rathore et al., 2008a, b; Rathore et al., 2009; Rathore et al., 2010; Westerberg et al., 2010). For instance, near real-time atline fluorescence spectroscopy has been applied to differentiate between misfolded and correctly folded proteins in order to make pooling decisions (Rathore et al., 2009). However, atline monitoring is time consuming, can lead to infrequent or delayed data and thus late detection of production irregularities, and is consequently rather inefficient for process control (van de Merbel et al., 1996).

An approach allowing frequent data collection is online HPLC and UHPLC. PAT tools based on this approach have been applied for a selective quantification of the product and co-eluting contaminants and have successfully been used to make real-time pooling decisions (Fahrner et al., 1998; Kaltenbrunner et al., 2012; Mendhe et al., 2014; Rathore et al., 2008a, b; Rathore et al., 2010). In addition to the expensive equipment, the major disadvantage of online HPLC or UHPLC is the rather long response time (usually several minutes) (van de Merbel et al., 1996). Although efforts have been made to decrease assay times (Fahrner et al., 1998; Kaltenbrunner et al., 2012; Mendhe et al., 2014; Rathore et al., 2008a, b; Rathore et al., 2010), the response time may require a slowdown of the process flow rate (Kaltenbrunner et al., 2012; Mendhe et al., 2014; Rathore et al., 2008a, b; Rathore et al., 2010) and/or the application of mathematical models to predict the protein titers in the upcoming fraction (Rathore et al., 2010, 2008a, b).

Next to online HPLC and UHPLC, PAT tools based on mathematical models in combination with univariate UV absorption or conductivity measurements have been proposed for inline monitoring and control of chromatography processes (Mendhe et al., 2014; Westerberg et al., 2010). Mendhe et al. could successfully establish a model based on the retention time of a characteristic peak eluting prior to the product peak. This model allowed product purity-based, real-time pooling decisions using univariate UV absorption measurements. The proposed PAT tool is easy to implement and does not require a modification of the chromatography system, but is not applicable for all separation issues as it requires a pre-peak as reporter. The PAT tool proposed by Westerberg et al. uses direct (conductivity of the loading buffer) and indirect (univariate UV absorption) measurements of critical process parameters that were identified during a chromatography model-based sensitivity analysis. However, the two presented case studies were based on simple separation issues. For real processes with many components, a chromatography model-based sensitivity analysis may not be easy to implement as it requires a prior model parameter determination for each component.

An alternative approach to differentiate between product and co-eluting contaminants that also allows a selective quantification of the species is spectral analysis in combination with Partial Least Squares Regression (PLS). Due to unequal number of the aromatic amino acid residues in different protein species, protein mid-UV absorption spectra vary and allow selective protein quantification in multicomponent mixtures using PLS. PLS has been successfully applied to calibrate models that correlate mid-UV absorption spectra with selective protein concentrations. The calibrated PLS models were successfully applied for a selective quantification by spectral analysis in various multicomponent protein mixtures. The

approach has also been successfully used as analytics for high throughput chromatography experiments (Hansen et al., 2011, 2013). The applicability of mid-UV absorption spectra in combination with PLS for selective protein quantification in multicomponent mixtures has also been confirmed by other researchers (Kamga et al., 2013). Recently, a powerful integration of software and chromatography hardware was introduced that enables the applicability of this approach for a selective inline quantification of co-eluting proteins in chromatography (Brestrich, 2014). This tool for selective inline quantification consists of a conventional AKTA[®] purifier liquid chromatography system, a diode array detector for the spectral measurement, a National Instruments data acquisitions device, and a software interface to Matlab[®]. The tool was rigorously compared with common offline analytics and found to deliver precise results for the retention times and peak flanks of a ternary model protein system. Further, the tool for selective inline quantification could be successfully applied for product purity-based, real-time pooling decisions.

So far, however, the tool was only applied for model proteins, as a Design of Experiments (DoE)-based PLS model calibration, requiring pure protein components, was used (Hansen et al., 2011, 2013). In complex applications with pharmaceutically relevant proteins, this would require an isolation of the target protein and all impurities. This is not only elaborate, but also impracticable regarding the material limitations during biopharmaceutical process development. Additionally, when used for inline applications, the DoE-based calibration method requires a time-consuming injection of each sample into the applied inline detector (Brestrich, 2014).

To overcome these limitations, a method was developed in this study that enables the PLS model calibration using recorded process data. The process data was generated by performing various chromatography runs with batch-to-batch variability and by recording the corresponding inline mid-UV absorption spectra. The selective protein concentrations of the elution peaks were determined using offline reference analytics based on collected fractions. In order to generate average spectra from the inline data and to correlate them with the reference analytics in a PLS model, a Visual Basic for Application (VBA) macro was developed. This process-data-based calibration method was established using a ternary model protein system consisting of lysozyme (lys), cytochrome c (cyt c), and ribonuclease A (rib A). Afterwards, the method was applied to calibrate PLS models for two different separation issues to demonstrate that the proposed tool for selective inline quantification (Brestrich et al., 2014) can be applied for the monitoring and control of real-life processes. In a first case study, models for a selective quantification of monoclonal antibody (mAb), its aggregates (HMWs—High Molecular Weights) and fragments (LMWs—Low Molecular Weights) were calibrated. The quantified species were co-eluting during a cation exchange (CEX) step in mAb purification. The second examined separation issue was the purification of transferrin (trf) and IgG from Cohn supernatant I using flow-through mode anion-exchange (AEX) chromatography. A detailed description of the process can be found in McCann et al. (2005). Besides from the co-eluting target molecules trf and IgG, the AEX flow-through was containing other, lower concentrated UV-absorbing species.

Materials and Methods

For both the method establishment and the two case studies, the general experimental procedure for the process-data-based model calibration was the same and is displayed in Figure 1: In the first step, various chromatography runs with batch-to-batch variability were performed. The elution peaks were afterwards deconvoluted by reference analytics based on collected fractions. Additionally, the mid-UV absorption spectra of the runs were recorded. For the method establishment and the first case study, this was realized using an inline diode array detector. The average spectra of each fraction were afterwards determined using a customized VBA macro. For the second case study, only the mid-UV absorption spectra of the fractions could be recorded as the study was performed at CSL Behring, Broadmeadows. The PLS models were finally generated by correlating the results of the fraction analytics with the corresponding spectra.

Method Establishment Using Model Proteins

Proteins and Buffers

Lys from hen egg white, rib A from bovine pancreas, and cyt c from equine heart were purchased from Sigma–Aldrich, St. Louis, MO. This system was chosen due to the stability of the proteins in

aqueous solutions regarding the formation of multimers and due to their advantageous isoelectric point distribution, allowing separation at neutral pH by cation exchange chromatography. Cyt c was declared as “product” and the two other proteins as “contaminants”. For the chromatography runs, the loading buffer was 20 mM sodium phosphate (pH 7). Elution was performed using a high-salt buffer consisting of 20 mM sodium phosphate as well as 500 mM sodium chloride (pH 7). For the reference analytics (analytical CEX chromatography), the loading buffer was 25 mM Tris (pH 8) and elution was performed using 25 mM Tris with 600 mM sodium chloride (pH 8). All buffer components were purchased from VWR, West Chester, PA. Buffers were 0.22 μm filtered and degassed by sonification before usage.

Chromatographic Instrumentation

The chromatography runs were performed using a specific experimental setup consisting of a conventional liquid chromatography system, a diode array detector, and a software interface to Matlab[®]. The liquid chromatography system was an AKTA[®] purifier 10 equipped with pump P-900, sample pump P-960, UV monitor UV-900 (10 mM optical path length), conductivity monitor C-900, pH monitor pH-900, autosampler A-905, and fraction collector Frac-950 (all GE Healthcare, Chalfont St. Giles,

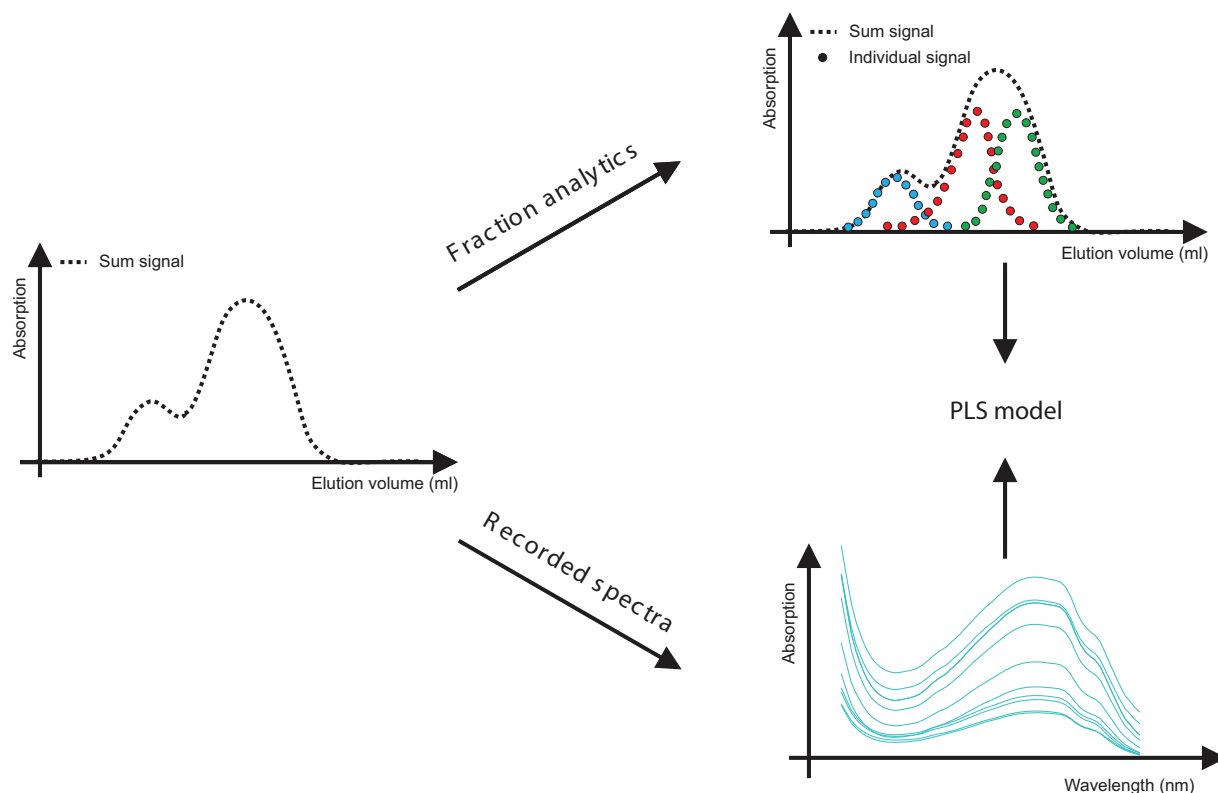


Figure 1. Process-data-based model calibration: Various chromatography runs with batch-to-batch variability were performed. The elution peaks were afterwards deconvoluted by reference analytics based on collected fractions. Additionally, the mid-UV absorption spectra of the runs or of collected fractions were recorded. For the PLS model calibration, the results of the reference analytics were finally correlated with the corresponding spectra.

UK). The instrumentation was controlled with UNICORN 5.31 software. In order to obtain inline mid-UV absorption spectra, a Dionex UltiMate 3000 Diode Array Detector (DAD) equipped with a semi-preparative flow cell (0.4 mm optical path length) and operated with Chromeleon 6.80 software (Thermo Fisher Scientific, Waltham, MA) was connected to the AKTA[®] Purifier 10. The data acquisition of the DAD was triggered using a customized software interface. A detailed description can be found in Brestrich et al. (2014).

Chromatography Runs

A HiTrap 1 mL column prepacked with SP Sepharose FF (GE Healthcare) was first equilibrated with 5 column volumes (CVs) of loading buffer and then loaded with 0.4 mg of the “product” and 0.2 mg of each “contaminant.” Afterwards, a wash of 5 CVs loading buffer was performed and the proteins were then eluted by various linear gradients from 0 to 500 mM sodium chloride. The gradient length was 1, 3, 5, and 7 CVs. The variable gradient length was performed to imitate batch-to-batch variability in a model protein study. Variable gradients lead to different resolutions and thus mixing ratios of the proteins as well as to different peak heights and thus maximal concentrations of the species. At the start of the gradient, the DAD was triggered to start data acquisition and recorded protein mid-UV absorption spectra in the band of 240–300 nm with 1 nm resolution. After the gradient, the column was regenerated for 5 CVs using elution buffer. The flow rate was 0.2 mL/min for all steps and experiments. The elution peaks were collected in deep well plates (VWR) in 150 μ L fractions. Afterwards, they were transferred to half area plates (Greiner BioOne, Kremsmünster, Austria) for the reference analytics. This procedure allowed an appropriate fraction collection as well as the required sample height in the wells for the reference analytics. As the reference analytics was based on collected fractions, protein concentrations that are averaged over the fraction size were determined. In contrast to that, the DAD recorded a continuous inline signal (3d field of the DAD). Therefore, spectra averaged over the fraction size were calculated from the recorded data before correlating them with the reference analytics by PLS. In order to generate these average spectra, a VBA macro was developed. The macro uses classes of the Chromeleon Software Developer Kit (Thermo Fisher Scientific) to export the recorded spectra of each fraction to Matlab[®]. The average spectrum of each fraction was then determined by Matlab[®].

Reference Analytics

As reference analytics, analytical CEX chromatography was performed with the collected fractions, using a Dionex UltiMate 3000 liquid chromatography system (Thermo Fisher Scientific). The system was composed of a HPG-3400RS pump, a WPS-3000TFC-analytical autosampler, a TCC-3000RS column thermostat, and a DAD3000RS detector. For each fraction, a Proswift SCX-1S 4.6 $\times$ 50 mm column (Thermo Fisher Scientific) was first equilibrated for 2.7 min with loading buffer, loaded with 20 μ L sample, washed for 1 min with loading buffer and then eluted by multiple salt steps: 20% elution buffer for 0.9 min, 75% elution

buffer for 0.7 min, and 100% elution buffer for 6.4 min. The flow rate was 0.5 mL/min for all steps.

First Case Study: CEX Step for mAb Purification

Process Material and Buffers

mAb Protein A eluate, donated by Rentschler (Laupheim, Germany), was 0.22 μ m filtered. The mAb concentration based on a provided extinction coefficient was determined using a NanoDrop 2000 UV-Vis Spectrophotometer (Thermo Fisher Scientific). For the chromatography runs, the loading buffer was 10 mM sodium citrate (pH 4.1). Elution and regeneration was performed using 10 mM sodium citrate with 1000 mM sodium chloride (pH 4.1). For the reference analytics (size exclusion chromatography), the buffer was 200 mM potassium phosphate with 250 mM potassium chloride (pH 7). All buffer components were purchased from VWR, West Chester, PA. Buffers were 0.22 μ m filtered and degassed by sonification before usage.

Chromatographic Instrumentation

For the chromatographic instrumentation used, cf. the corresponding section in “Method establishment using model proteins”.

Chromatography Runs

A MediaScout 1 mL column prepacked with POROS HS 50 (Carlsbad, CA) was purchased from Atoll, Weingarten, Germany. The column was equilibrated with 5 CVs of loading buffer and then loaded with filtered protein A pool. Afterwards, a wash of 5 CVs loading buffer was performed and the elution was carried out using various linear gradients from 0 to 70% elution buffer. Gradient length, imitating the batch-to-batch variability, was 10, 20, and 30 CVs. At the start of the gradient, the DAD was triggered to start data acquisition and recorded protein absorption spectra in the band of 240–410 nm with 1 nm resolution. Finally, the column was regenerated for 5 CVs using 100% elution buffer and was then stripped with 0.5 M sodium hydroxide over night. The flow rate was 0.2 mL/min for all steps and experiments. The elution peaks were collected in 150 μ L fractions as described in the section “Method Establishment Using Model Proteins”. Further, the average mid-UV absorption spectra were determined as explained in the section “Method Establishment Using Model Proteins”.

Reference Analytics

As reference analytics, the total mAb concentration of each fraction was first of all determined using the NanoDrop 2000. Afterwards, size exclusion chromatography (SEC) was performed with the collected fractions to determine the mass fractions of mAb, HMWs, and LMW using the same Dionex UltiMate 3000 liquid chromatography system as described in the section “Method Establishment Using Model Proteins”. For each fraction, 20 μ L sample were loaded on a 4.6 $\times$ 150 mm TSKgel SuperSW mAb HTP column (Tosoh, Tokyo, Japan). In order to reduce the assay time, interlaced injections were used (Diederich et al., 2011).

Second Case Study: AEX Step in Protein Purification From Cohn Supernatant I

Process Material and Buffers

Lipid- and euglobulin-depleted Cohn supernatant I was taken directly from the full scale process as an intermediate sample and was 0.22 μm filtered. The major components were albumin (approx. 73%), IgG (approx. 19%), and trf (approx. 5%). Other minor components were IgA and IgM. For the chromatography runs, the loading buffer used was sodium acetate at pH 5.2. Elution was performed using sodium acetate at pH 4.5. All buffers were 0.22 μm filtered before usage.

Chromatographic Instrumentation

The chromatography runs were performed using an AKTA[®] explorer 100 system equipped with pump P-900, sample pump P-960, UV monitor UV-900 (10 mm optical path length), conductivity monitor C-900, pH monitor pH-900, and fraction collector Frac-950 (all GE Healthcare). The instrumentation was controlled with UNICORN 5.31 software.

Chromatography Runs

An XK16/40 column (GE Healthcare) was packed to a height of 17.5 cm using DEAE Sepharose FF resin by GE Healthcare, and the quality of the packed column was assessed by measuring HETP. A packed column was rejected if the HETP came out to be >0.06 cm. The column was equilibrated using loading buffer. A sample of lipid- and euglobulin-depleted supernatant I was loaded onto the column at 65 g/L. Loosely bound protein was flushed off the column using loading buffer. The flow-through was collected in fractions of 5 mL. The fractions were then analyzed by measuring their individual absorbance spectra between 240 nm and 300 nm using a Cary 50 Spectrophotometer by Agilent Technologies (Santa Clara, CA) and a quartz glass cuvette. Finally, bound protein was eluted using the corresponding buffer. The flow rate was 3.33 mL/min for all steps and experiments.

Reference Analytics

The levels of plasma proteins were measured by nephelometry using an Immage 800 Immunochemistry system by Beckman-Coulter (Pasadena, CA). The Immage system was calibrated using standard samples available from Beckman-Coulter.

Partial Least Squares Regression

In order to correlate the average mid-UV absorption spectra with the selective protein concentrations Partial Least Squares Regression (PLS) technique was applied (Eriksson et al., 2006; Höskuldsson, 1988; Martens and Næs, 1989). PLS aims to reduce data by only applying significant variance in data for correlating several input variables with variables of interest. To achieve a data reduction, variables containing similar information are summarized in so-called latent variables (LVs). Thus, the number of LVs expresses a measure of data reduction, where a low number of LVs

corresponds to a high reduction. The information content included in each LV is high for the first LV and decreases for the following LVs. This means that the variance in the data can be explained by only a few LVs, leaving out higher LVs that mainly contain detector noise. The higher the number of LVs used in a PLS model, the more precise is the established model for chromatography runs used in the calibration. However, the use of too many latent variables can result in overfitting. Therefore, the thorough examination of the number of LVs used in a PLS model is crucial for its prediction precision. In order to determine a reasonable number of LVs, the root mean square error for the prediction (RMSEP) in dependence on the number of LVs used in the calibration was calculated for a validation run (Eriksson et al., 2006).

The PLS model calibration based on the obtained average mid-UV absorption spectra and concentrations from the corresponding reference analytics was performed using SIMCA (Umetrics, Umeå, Sweden). The software was used because of the easiness of data preprocessing and visualization. SIMCA applies the NIPALS-algorithm for PLS. Before performing PLS, spectra were preprocessed by mean centering using SIMCA. However, it is also possible to perform the PLS regression using Matlab[®] or other available software. After their calibration, the PLS models were applied to calculate protein concentrations from inline mid-UV absorption spectra.

Results and Discussion

An application of the tool for selective inline quantification (Brestrich, 2014) towards real-life separation issues demands for a fast and reliable calibration of PLS models. The calibration method should not require pure protein components and thus an elaborate isolation of the target protein and all impurities. This work demonstrates that process data can be successfully applied to calibrate PLS models for the inline monitoring of co-eluting proteins in chromatography. In order to generate average spectra of the recorded process data, a VBA macro was successfully developed. The methodology was established using model proteins. Afterwards, it was demonstrated in two case studies that the process-data-based calibration is applicable towards real-life separation issues.

Method Establishment Using Model Proteins

In order to imitate batch-to-batch variability in a model protein study, the gradient length was varied. This resulted in different resolutions and thus mixing ratios of the model proteins as well as different peak heights and thus maximal concentrations of the species. For the PLS model calibration, the results of the fraction analytics by analytical CEX chromatography for the 1, 3, and 7 CVs gradient run were correlated with the corresponding average mid-UV absorption spectra using PLS. Eight PLS models with 1–8 LVs were calibrated. The PLS models were afterwards applied to predict the selective protein concentrations of the run with a 5 CV gradient length from the average mid-UV absorption spectra. The RMSEP between the predicted concentrations and the results from the reference analytics with respect to the number of LVs used in the calibration is displayed in Figure 2A. As to be expected, the RMSEP

was first decreasing with increasing number of LVs for all proteins and then slightly increasing at certain number of LVs when overfitting was reached. The best combination of low RMSEP and low number of LVs was given at 6 LVs for rib A, at 5 LVs cyt c, and at 7 LVs for lys. As this combination was different for each protein, an individual PLS model for each protein using this determined optimal number of LVs was calibrated using the 1, 3, and 7 CVs gradient run. The RMSEP of the final model was 0.008 g/L for rib A, 0.016 g/L for cyt c, and 0.005 g/L for lys.

The calibrated PLS models were afterwards applied for a peak deconvolution of the validation run with a 5 CV gradient using inline mid-UV absorption spectra. This means that in contrast to the PLS model calibration, where average spectra of each fraction were applied, the continuous inline signal (3d field of the DAD) was used. The application of PLS models generated by absorption spectra averaged over fraction size to evaluate inline absorption spectra was possible due to high resolution of the collected fractions. The resulting chromatogram of the inline peak deconvolution is displayed in Figure 2B (solid lines) and compared with the offline fraction analytics (dashed lines). A good agreement between the prediction by the PLS model and the reference analytics for both the peak maxima and peak flanks was observed. This clearly shows the precision of the models generated by the process-data-based calibration and the applicability of this calibration method towards inline monitoring of real-life separation issues.

Compared to the prediction of the PLS model calibrated by the DoE-method (Brestrich et al., 2014), where calibration samples from pure protein stock solutions are mixed according to a DoE, a higher precision at the peak maxima was achieved. This might be explained by the fact that using a DoE-based calibration, all

possible mixing ratios and concentration levels of the model proteins are accounted for using as few calibration samples as possible. This results in PLS models that are valid for a large concentration level and mixing ratio space, but which are not specialized for the mixing ratios and concentration levels occurring in the chromatography run to be monitored. In contrast to that, PLS models calibrated with process data exhibit a high precision within the process window they were calibrated for but decrease in their prediction precision outside of this window.

One possibility to display this process window is plotting the scores of the calibration samples on all LVs used in a model. Exemplarily, Figure 3A displays the scores of the calibration samples (1, 3, and 7 CVs gradient) for the cyt c model on the first and second LV, which together explained a major part (88%) of the variance in the cyt c concentration. Additionally, the scores of the 5 CV gradient validation run are displayed. As they are located within the process window, the prediction precision was high for cyt c in the validation run (cf. Fig. 2B). Obviously, this also has to be examined for the higher LVs (data not shown). Furthermore, it can be seen in Figure 3A that a model calibrated using the 1, 5, and 7 CVs gradient run can precisely predict the 3 CV gradient run (cf. Fig. 3B). In contrast to that, it is neither possible to apply the 1 CV nor the 7 CV gradient run as a validation run, as they would then lie outside of the process window spread out by the calibration samples (cf. Fig. 3A). The prediction of one of these two runs by using the other ones for the model calibration equals an extrapolation of the corresponding PLS model for some concentrations and mixing ratios. This leads to a lower prediction precision (cf. Fig. 3C–D). Consequently, all batch-to-batch variability that might occur in a process should be accounted for in the corresponding models. The required data for this purpose might be obtained from process

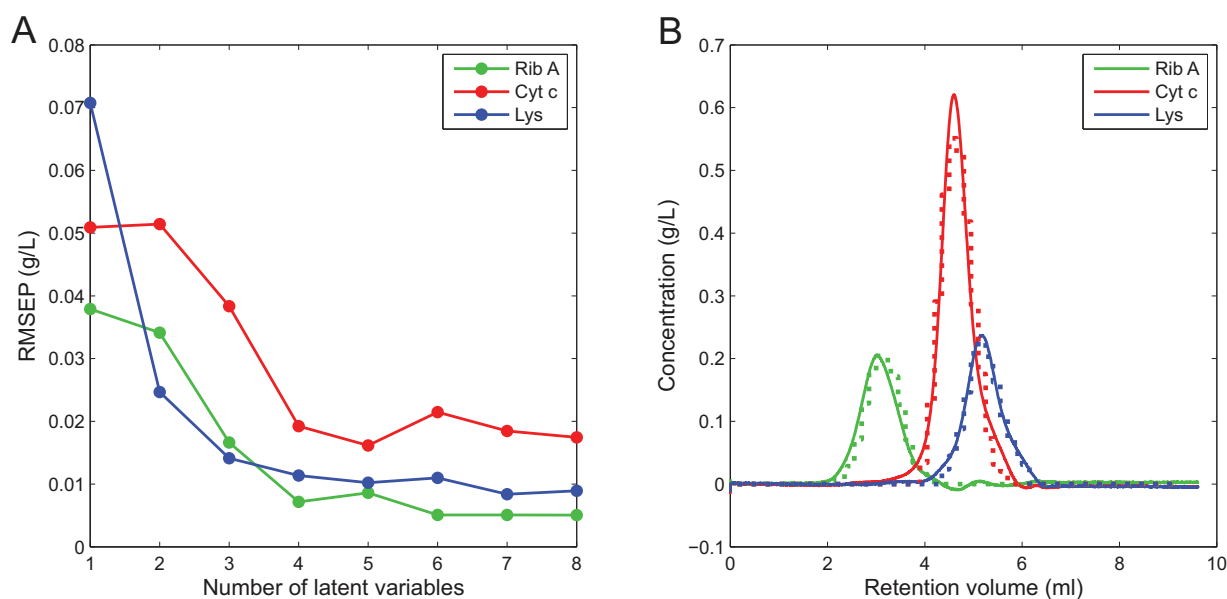


Figure 2. A: Calculated root mean square error for the prediction (RMSEP) of rib A, cyt c, and lys in the validation run with respect to the number of latent variables used for the PLS model calibration. B: Resulting chromatogram of the inline peak deconvolution for the validation run (solid lines) and offline reference analytics displayed as stairs (dashed lines).

development and robustness studies in order to minimize the experimental effort.

First Case Study: CEX Step for mAb Purification

In mAb platform processes, the protein A capture and low pH virus inactivation is usually followed by two polishing chromatography steps aiming to reduce remaining host cell proteins, HMWs, LMWs, DNA, and leached Protein A (Shukla et al., 2007). The CEX step typically aims to reduce HMWs and LMWs. These two species exhibit similar physico-chemical properties as the product, which often results in a poor resolution. As the pooling in this purification step is typically based on univariate UV absorption, batch-to-batch variability results in inconsistent pool purities. The goal of this study was to imitate batch-to-batch variability by different gradient lengths and to calibrate PLS models for the inline quantification of mAb, HMWs, and LMWs using this process data. Based on the inline quantification by the PLS models, product-purity based, real-time pooling decisions would be possible.

For the model calibration, the results of the fraction analytics by SEC for the 10 and 30 CVs gradient run were correlated with the corresponding average mid-UV absorption spectra using PLS. Ten PLS models with 1–10 LVs were calibrated. The PLS models were afterwards applied to predict the selective concentrations of monomer, HMW1, HMW2, and LMW in the run with a 20 CV gradient length from the average mid-UV absorption spectra. The RMSEP between prediction and reference analytics with respect to the number of LVs used in the calibration is displayed in Figure 4. The best combination of low RMSEP and low number of LVs was given at 6 LVs for monomer, at 7 LVs for HMW 1, at 2 LVs for LMW, and at 7 LVs for HMW 2. As for the model proteins, individual PLS models using this determined optimal number of LVs were calibrated using the 10 and 30 CVs gradient run. The RMSEP of the final model was 0.014 g/L for monomer, 0.0036 g/L for HMW 1, 0.0019 g/L for LMW, and 0.0007 g/L for HMW 2.

The calibrated PLS models were applied for an inline peak deconvolution of the validation run with a 20 CV gradient. The resulting chromatograms are displayed in Figure 5 (blue lines) and

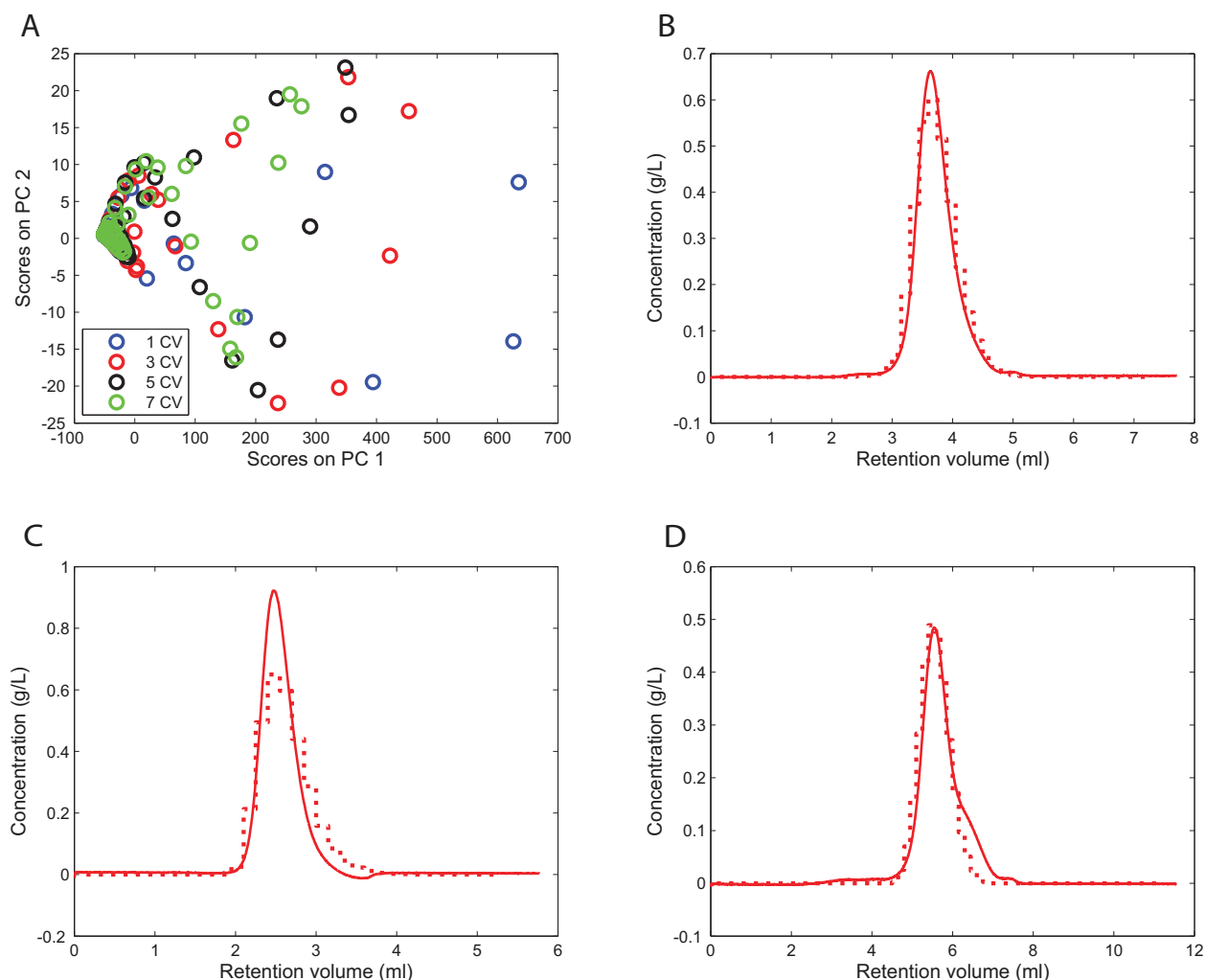


Figure 3. A: Scores on the first and second LV for the cyt c model. B–D: Resulting chromatogram of the inline peak deconvolution of cyt c (solid lines) for a 3 CV (B), 1 CV (C), or 7 CV (D) gradient length validation run and offline reference analytics displayed as stairs (dashed lines). The remaining samples were applied as calibration samples.

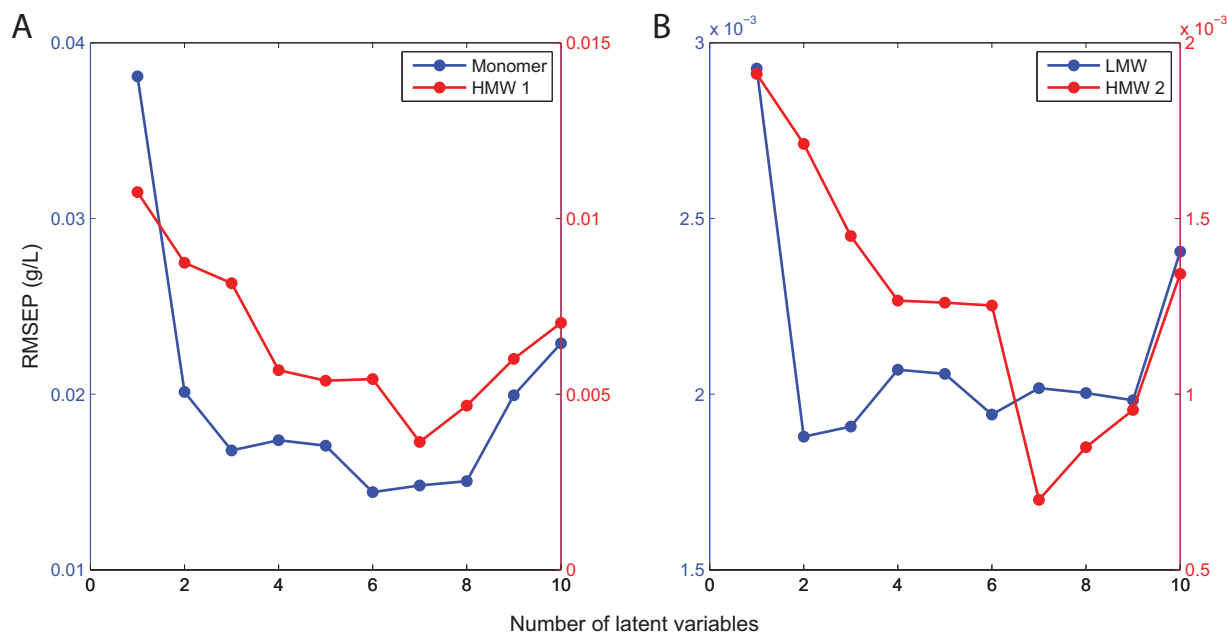


Figure 4. **A:** Calculated root mean square error for the prediction (RMSEP) of monomer and HMW 1 in the validation run with respect to the number of latent variables used for the PLS model calibration. **B:** RMSEP of LMW and HMW 2 in the validation run with respect to the number of latent variables used for the PLS model calibration.

compared with the offline fraction analytics (red stair functions). A good agreement between the prediction of the PLS models and the reference analytics was observed for all four monitored species. The agreement between prediction and reference for the aggregate species (HMW 1 and 2) was found to be slightly lower. This might be explained with a lower precision of the reference analytics for these species, as no complete baseline separation could be achieved during the SEC. But the results are acceptable regarding the fact that the overall trend, especially the peak flanks, could be predicted.

Based on these findings, the question on the origin of spectral differences between monomers and aggregates arises. An effect based on molecular structure might be conformational changes during the aggregation process which lead to changes in the microenvironment around the aromatic amino acid residues (Donovan, 1969; Kuelto et al., 2003; Ragone et al., 1984). More recently, Thakkar et al. (Thakkar et al., 2012) found out that even protein–protein interactions in highly concentrated solutions can lead to changes in the micro-environment around the aromatic amino acid residues and thus to changes in UV absorbance. With larger aggregates, these structural effects might be overruled by scattering effects. Scattering of incident light on solute protein multimers prevents the light from reaching the detector and results in an artificial increment of the absorbance values (Jiskoot and Crommelin, 2005). Usually, the scattering artefact is corrected. However in this study, it was desired. The spectral differences between LMWs and the other species can be ascribed to differences in the number and ratio of the aromatic amino acid residues.

Second Case Study: AEX Step in Protein Purification From Cohn Supernatant I

In contrast to the first case study and the method establishment using model proteins, the mid-UV absorption spectra of the second case study were obtained from an offline spectrophotometer as the study was performed at CSL Behring, Broadmeadows. For this study, the mid-UV absorption spectra of collected fractions were recorded. Consequently, the calibrated model cannot be used for an inline monitoring as it is specific for the applied detector. However, process data was used for the PLS model calibration and the developed procedure to generate average spectra from inline data could be easily transferred to this study.

At CSL Behring, DEAE Sepharose FF chromatography is used to purify albumin, IgG, and trf from Cohn supernatant I. Therefore, supernatant I is loaded at pH 5.2, allowing albumin (pI 4.8) to bind to the resin while IgG (pI 4.35–9.95) passes the column unretained (Prin et al., 1995). Other proteins in Cohn supernatant I such as trf (pI 5.0–6.0), IgA (pI 4.0–7.1), and IgM (pI 4.0–9.1) are partially retained and therefore present in both the IgG- and the albumin-rich fraction (Morgan, 1981; Prin et al., 1995). The DEAE flow-through is pooled in two fractions in order to gain both trf and IgG. However, the resolution of both proteins is strongly dependent on the sample loading, protein concentration in supernatant I, as well as on the conductivity of supernatant I and of the equilibration buffer (McCann et al., 2005). Consequently, the main goal for the PLS model calibration was the deconvolution of co-eluting trf and IgG in the flow-through in order to enable real-time pooling of these two target

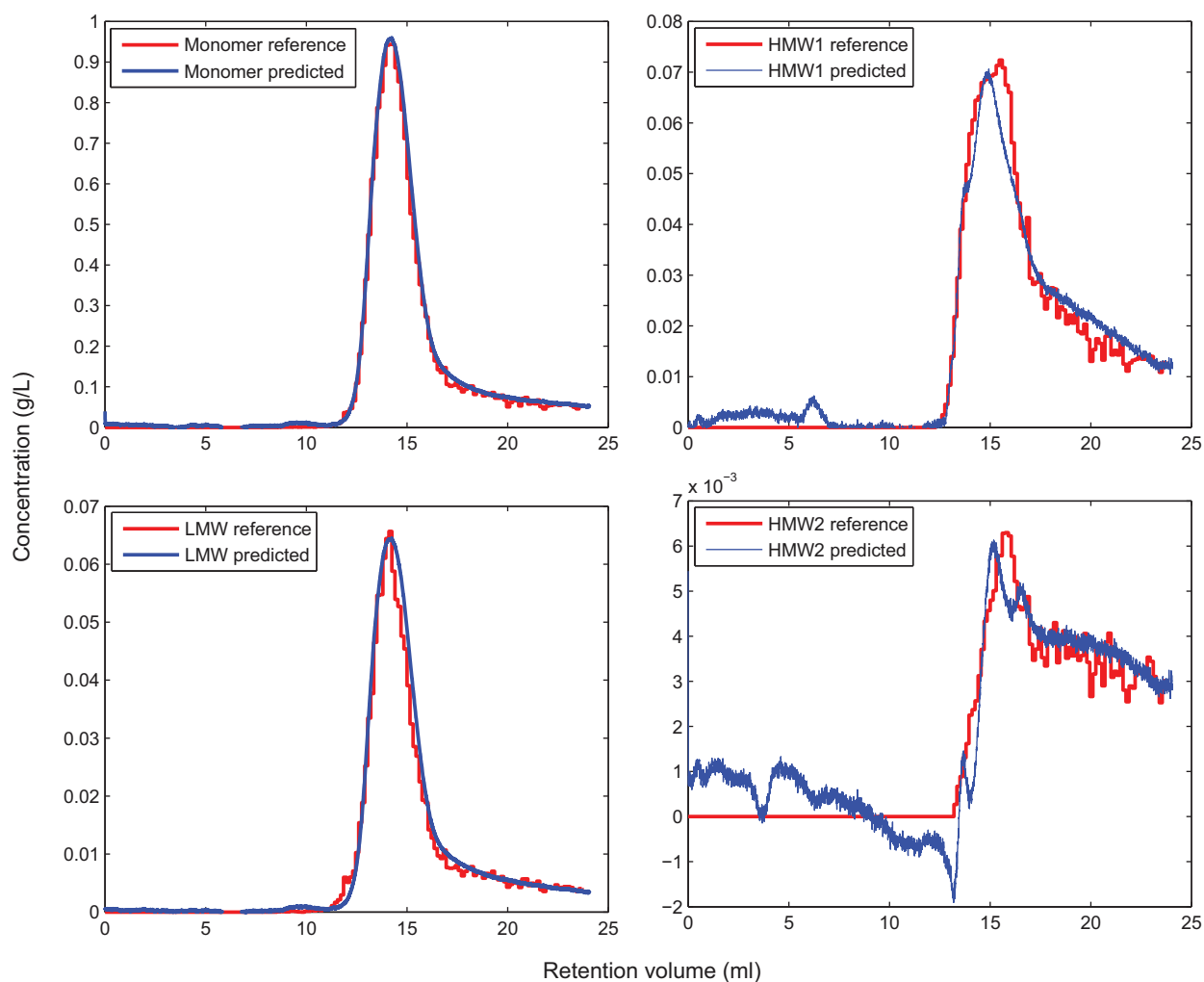


Figure 5. Resulting chromatograms of the inline peak deconvolution for monomer, LMW, HMW1, and HMW2 in the validation run as well as the resulting offline reference analytics. The results of the reference analytics (SEC chromatography) based on collected fractions are displayed as stairs.

proteins. Nevertheless, IgM and IgA, which are separated at a later point of the process, were included in the model as well.

The standard process exhibited a batch-to-batch variability as displayed by the trf elution profiles in Figure 6A. Therefore, two of these runs were applied for the model calibration and a third run was used as validation run. Six PLS models with 1–6 LVs were calibrated and then afterwards applied to predict the selective protein concentrations of IgG, trf, IgM, and IgA of the validation run. Figure 6B shows the determined RMSEP between prediction and reference analytics with respect to the number of LVs used in the calibration. The best combination of low RMSEP and low number of LVs was given at 5 LVs for all proteins examined. Consequently, only one PLS model with 5 LVs was applied. The RMSEP of this model was 0.305 g/L for IgG, 0.137 g/L for trf, 0.011 g/L for IgM, and 0.049 g/L for IgA.

The calibrated PLS model was used to predict the protein concentrations of the validation run from the average mid-UV

absorption spectra of the collected fractions. The results for all four proteins examined are displayed in Figure 7 and compared with the corresponding reference analytics. A good prediction of the elution profiles for the higher concentrated target proteins IgG and trf could be achieved. Additionally, a good agreement between prediction and reference of the lower concentrated IgM was observed. Solely for IgA differences in the total concentrations were found indicating a lower precision of the model for this protein. But the overall trend of the elution profile could be predicted. The cause was, however, not further pursued because the goal of the study was the accurate prediction of the IgG and trf as this has the greatest impact on process efficiency and IgG recovery. IgA is a minor constituent of the crude IgG and does not have a significant impact on the downstream processing of the IgG fraction. However, if the exact determination of IgA was important for the process, more calibration runs would be needed in order to calibrate a model that can predict IgA more precisely.

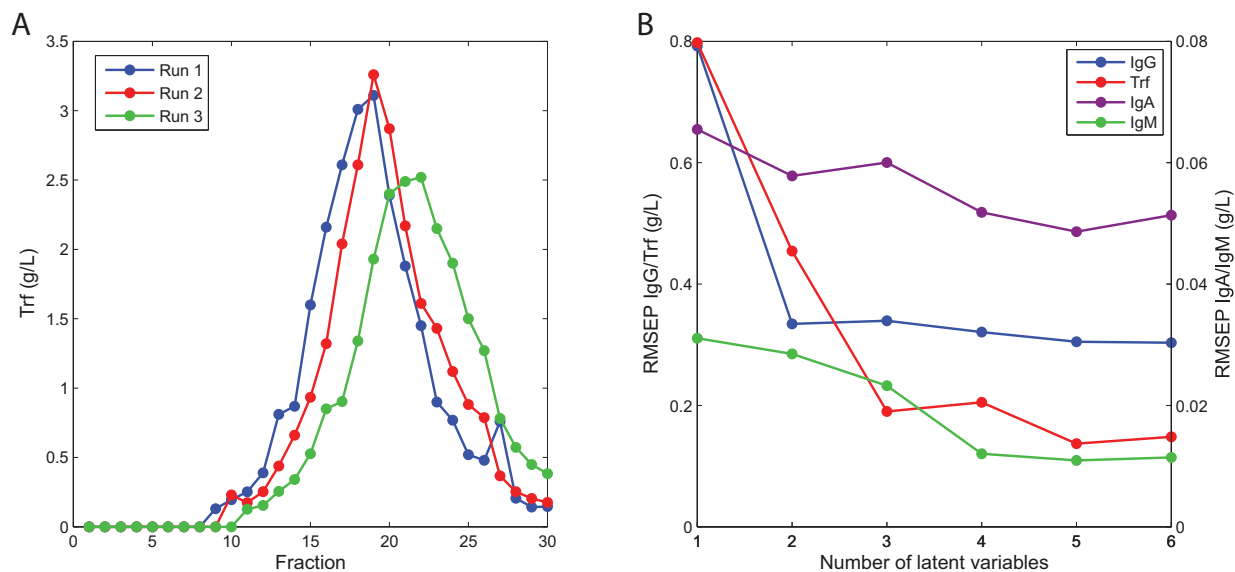


Figure 6. **A:** Batch-to-batch variability in the trf elution profiles of the three runs with a standard resin load of 65 g/L. **B:** Calculated root mean square error for the prediction (RMSEP) of IgG, trf, IgA, and IgM in the validation run with a resin load of 65 g/L.

Conclusion and Outlook

The use of inline mid-UV absorption spectra in combination with PLS for a real-time monitoring of co-eluting proteins in chromatography demands for a fast and reliable calibration method of PLS models, which does not require pure proteins. In this paper, a process-data-based PLS model calibration was successfully developed. The process data was generated by performing various

chromatography runs with batch-to-batch variability and recording the corresponding inline mid-UV absorption spectra. The elution peaks were deconvoluted by offline reference analytics based on collected fractions. In order to generate average spectra from the recorded inline data, a VBA Macro was developed. The average mid-UV absorption spectra and the reference analytics were correlated in various PLS models and successfully applied for inline peak deconvolutions of co-eluting proteins in validation runs. The

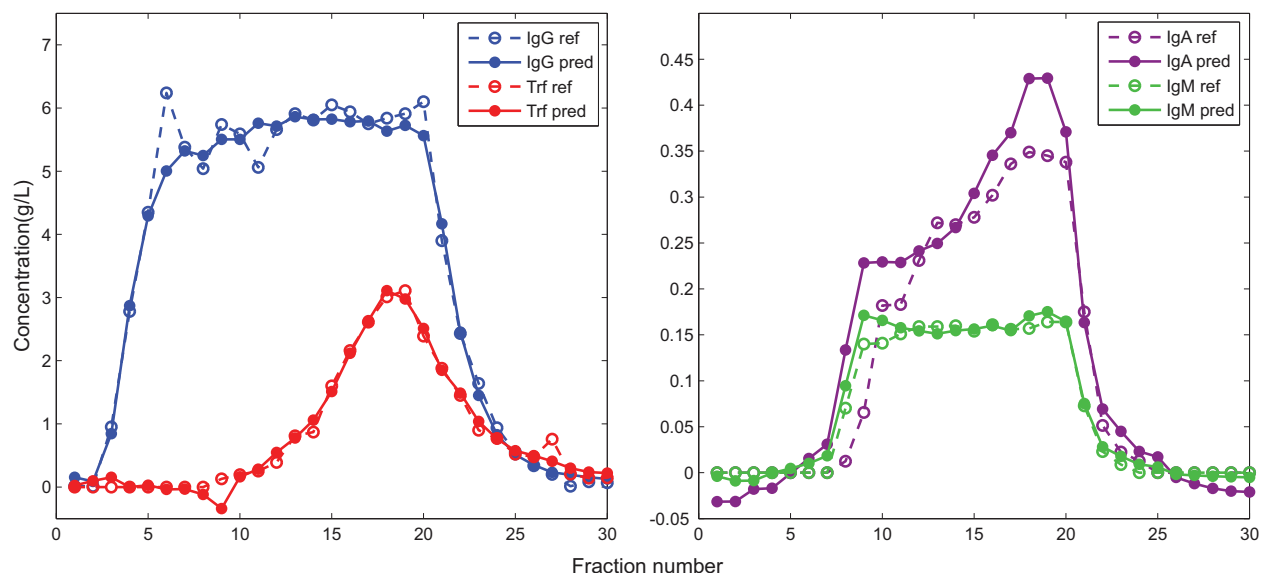


Figure 7. PLS model prediction based on collected fractions and reference analytics for IgG, trf, IgA, and IgM in the validation run.

process-data-based calibration method was found to deliver precise results within the calibrated process window. As to be expected, the prediction precision outside this window decreased. Consequently, the final variability of a process should be accounted for in a corresponding PLS model. Suitable data for this could be obtained from process robustness studies. The process-data-based calibration method was established using a ternary model protein system and could successfully be transferred on real-life separation issues. Consequently, the presented calibration method together with the proposed tool for selective inline quantification (Brestrich et al., 2014) has a great potential for the real-time monitoring of future chromatography steps. This approach offers a tool to manage batch-to-batch variability by real-time pooling and to enable consistent pool purities, which is one of the key requirements within the FDA's PAT concept. Future work will focus on monitoring of HCPs and on the applicability of the presented approach for completely loaded columns.

This work was supported by a grant from the Ministry of Science, Research, and the Arts of Baden-Württemberg, Germany (Az.: 33-7533-7-11.6-2). We kindly thank Rentschler (Laupheim, Germany) for donating Protein A pool.

References

- Brestrich N, Briskot T, Osberghaus A, Hubbuch J. 2014. A tool for selective inline quantification of co-eluting proteins in chromatography using spectral analysis and partial least squares regression. *Biotechnol Bioeng* 111:1365–1373.
- Diederich P, Hansen SK, Oelmeier SA, Stolzenberger B, Hubbuch J. 2011. A sub-two minutes method for monoclonal antibody-aggregate quantification using parallel interlaced size exclusion high performance liquid chromatography. *J Chromatogr A* 1218:9010–9018.
- Donovan JW. 1969. Chemistry and metabolism of macromolecules: Changes in ultraviolet absorption produced by alteration of protein conformation. *J Biol Chem* 244:1961–1967.
- Eriksson L, Johansson E, Kettaneh-Wold N, Trygg J, Wikström C, Wold S. 2006. Multi- and megavariate data analysis. Umeå: Umetrics Academy. p 95.
- Fahrner RL, Lester PM, Blank GS, Reifsnnyder DH. 1998. Real-time control of purified product collection during chromatography of recombinant human insulin-like growth factor-I using an on-line assay. *J Chromatogr A* 827:37–43.
- FDA. 2004. Guidance for Industry. PAT—A framework for innovative pharmaceutical development, manufacturing and quality assurance.
- Follman DK, Fahrner RL. 2004. Factorial screening of antibody purification processes using three chromatography steps without protein A. *J Chromatogr A* 1024:79–85.
- Hansen SK, Jamali B, Hubbuch J. 2013. Selective high throughput protein quantification based on UV absorption spectra. *Biotechnol Bioeng* 110:448–460.
- Hansen SK, Skibsted E, Staby A, Hubbuch J. 2011. A label-free methodology for selective protein quantification by means of absorption measurements. *Biotechnol Bioeng* 108:2661–2669.
- Höskuldsson A. 1988. PLS regression methods. *J Chemom* 2:211–228.
- Jiskoot W, Crommelin D. 2005. Methods for structural analysis of protein pharmaceuticals. Ar-lington: AAPS Press. p 9.
- Kaltenbrunner O, Lu Y, Sharma A, Lawson K, Tressel T. 2012. Risk-benefit evaluation of on-line high-performance liquid chromatography analysis for pooling decisions in large-scale chromatography. *J Chromatogr A* 1241:37–45.
- Kamga MH, Lee WH, Liu J, Yoon S. 2013. Rapid quantification of protein mixture in chromatographic separation using multi-wavelength UV spectra. *Biotechnol Prog* 29:664–671.
- Kueltzo LA, Ersoy B, Ralston JP, Middaugh CR. 2003. Derivative absorbance spectroscopy and protein phase diagrams as tools for comprehensive protein characterization: A bGCSF case study. *J Pharm Sci* 92:1805–1820.
- Martens H, Næs T. 1989. Multivariate calibration. New York: Wiley.
- McCann KB, Hughes B, Wu J, Bertolini J, Gomme PT. 2005. Purification of transferrin from Cohn supernatant I using ion-exchange chromatography. *Biotechnol Appl Biochem* 42:211–217.
- Mendhe R, Muralikrishnan T, Patil N, Rathore AS. 2014. Comparison of PAT based approaches for making real-time pooling decisions for process chromatography—Use of feed forward control. *J Chem Technol Biotechnol* 90(2):341–348.
- Morgan EH. 1981. Transferrin, biochemistry, physiology and clinical significance. *Mol Aspects Med* 4:1–123.
- Prin C, Bene MC, Gobert B, Montagne P, Faure GC. 1995. Isoelectric restriction of human immunoglobulin isotypes. *Biochim Biophys Acta* 1243:290–298.
- Ragone R, Colonna G, Balestrieri C, Servillo L, Irace G. 1984. Determination of tyrosine exposure in proteins by second-derivative spectroscopy. *Biochemistry* 23:1871–1875.
- Rathore AS, Li X, Bartkowski W, Sharma A, Lu Y. 2009. Case study and application of process analytical technology (PAT) towards bioprocessing: Use of tryptophan fluorescence as at-line tool for making pooling decisions for process chromatography. *Biotechnol Prog* 25:1433–1439.
- Rathore AS, Parr L, Dermawan S, Lawson K, Lu Y. 2010. Large scale demonstration of a process analytical technology application in bioprocessing: Use of on-line high performance liquid chromatography for making real time pooling decisions for process chromatography. *Biotechnol Prog* 26:448–457.
- Rathore AS, Wood R, Sharma A, Dermawan S. 2008a. Case study and application of process analytical technology (PAT) towards bioprocessing: II. Use of ultra-performance liquid chromatography (UPLC) for making real-time pooling decisions for process chromatography. *Biotechnol Bioeng* 101:1366–1374.
- Rathore AS, Yu M, Yeboah S, Sharma A. 2008b. Case study and application of process analytical technology (PAT) towards bioprocessing: Use of on-line high-performance liquid chromatography (HPLC) for making real-time pooling decisions for process chromatography. *Biotechnol Bioeng* 100:306–316.
- Shukla A, Hubbard B, Tressel T, Guhan S D L. 2007. Downstream processing of monoclonal antibodies: Application of platform approaches. *J Chromatogr B* 848:28–39.
- Thakkar SV, Allegrè KM, Joshi SB, Volkin DB, Middaugh CR. 2012. An application of ultraviolet spectroscopy to study interactions in proteins solutions at high concentrations. *J Pharm Sci* 101:3051–3061.
- van de Merbel NC, Lingeman H, Brinkman UAT. 1996. Sampling and analytical strategies in on-line bioprocess monitoring and control. *J Chromatogr A* 725:13–27.
- Westerberg K, Degerman M, Nilsson B. 2010. Pooling control in variable preparative chromatography processes. *Bioprocess Biosyst Eng* 33:375–382.
- Zhou JX, Dermawan S, Solamo F, Flynn G, Stenson R, Tressel T, Guhan S. 2007. PH-conductivity hybrid gradient cation-exchange chromatography for process-scale monoclonal antibody purification. *J Chromatogr A* 1175:69–80.